

Task 5: Reflection and Video Presentation
CSC 3114: Cryptology and Coding Theory,
Block Ciphers, AES and Integrity

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Video Presentation

Link: <https://drive.google.com/file/d/1TARreFrYcpibNIpK1Uo6iej6iJCPtoCY/view?usp=sharing>

Reflection: The Most Important Security Lesson

The experiments demonstrated that the security of a cryptographic system depends not only on the strength of the algorithm, but also on how it is implemented and used. The AES avalanche experiment showed strong diffusion, with changing one bit of the plaintext resulting in an average change of 49.9% of the ciphertext bits, which is close to the ideal 50%.

The AES modes experiment further demonstrated that both CBC and CTR can successfully encrypt and decrypt data when used correctly. Fresh IVs and nonces ensured that encrypting the same message did not produce the same ciphertext. However, the experiment also highlighted the risks of incorrect implementation. In particular, reusing a CTR nonce with the same key can expose information about the plaintexts, while CBC and CTR do not provide authentication or protection against message modification.

The hashing experiment showed a similar distinction between confidentiality and integrity. A plain SHA-256 hash could be recalculated after a message was modified, whereas HMAC-SHA256 detected the modification because it requires a secret key.

Overall, the experiments showed that secure communication requires both confidentiality and integrity, together with proper key, IV, and nonce management. This explains the practical use of authenticated encryption methods such as AES-GCM.